Project BattleBot

Line-Following Track

Team IT-1D | Sub-Team BB008

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Line-Following Track – Setup and Description

The robotic system is built on an Arduino Nano platform and designed to operate on a curved line-following track. The setup integrates multiple sensors and actuators to enable autonomous navigation, object interaction, and real-time decision-making.

Hardware Components

The system consists of the following main components:

- 8 Analog Line Sensors (A0–A7)

  Used to detect the position of the line relative to the robot.
- Ultrasonic Sensor (TRIG: 7, ECHO: 8)

  Used for both start signal detection (flag) and obstacle detection.
- DC Motors (PWM control)
  - Left motor: pins 3 (back), 5 (forward)
  - Right motor: pins 11 (back), 6 (forward)

    Responsible for movement and steering.
- Servo Motor (Pin 9)

  Controls the gripper mechanism to grab and release objects.
- NeoPixel LEDs (Pin 12)

  Provide visual feedback for robot states (e.g., movement, turning, stopping).

Inputs (Sensors → Robot)

Name	Type	Pin	Description
Ultrasonic Trigger	Digital Output	7	Sends signal to measure distance
Ultrasonic Echo	Digital Input	8	Receives reflected signal
Line Sensor 0	Analog Input	A0	Left-most line detection
Line Sensor 1	Analog Input	A1	Line detection

Line Sensor 2	Analog Input	A2	Line detection
Line Sensor 3	Analog Input	A3	Line detection
Line Sensor 4	Analog Input	A4	Line detection
Line Sensor 5	Analog Input	A5	Line detection
Line Sensor 6	Analog Input	A6	Line detection
Line Sensor 7	Analog Input	A7	Right-most line detection

Logical Inputs (Software-Level)

Name	Source	Purpose
Flag Detection	Ultrasonic	Detect start condition
Obstacle Detection	Ultrasonic	Trigger avoidance
Line Position	Line sensors	Control steering
End Detection	Line sensors	Detect finish (all black)

Outputs (Robot → Actions)

Name	Type	Pin	Description
Motor Left Forward	PWM	5	Moves left wheel forward
Motor Left Backward	PWM	3	Moves left wheel backward
Motor Right Forward	PWM	6	Moves right wheel forward
Motor Right Backward	PWM	11	Moves right wheel backward
Servo Gripper	PWM	9	Opens/closes gripper
NeoPixel LED	Digital	12	Status lighting

Logical Outputs (Behavior)

Name	Action
driveForward()	Move robot forward
pivotLeft()	Turn left

pivotRight()	Turn right
driveBackward()	Reverse
openGripper()	Release object
closeGripper()	Grab object
LED feedback	Status indication

System Description

The robot follows an **Input–Process–Output architecture**, where sensor data is continuously processed to control movement and behavior.

1. Input Layer

The robot collects data from:

- Line sensors → detect line position
- Ultrasonic sensor → detect flag and obstacles

2. Processing Layer

The core logic is implemented using a **finite state machine (FSM)**:

States include:

- WAIT_FOR_FLAG → start condition
- DRIVE, TURN, LINE_FOLLOW → navigation
- AVOID_* → obstacle avoidance
- STOPPED, BACKWARD, OPEN → end sequence

The system also uses:

- **Weighted sensor calculation** to determine position
- **Error-based correction** for steering
- **Timing control (millis)** for non-blocking actions

3. Output Layer

Based on decisions, the robot controls:

- Motors → movement and direction
- Servo → object handling
- LEDs → status indication

Behavior Flow

The robot follows a structured sequence of actions:

1. **Start Detection**

   The robot waits until the flag is removed using ultrasonic sensing.

2. **Object Interaction**

   The robot opens its gripper, moves forward, and grabs an object.

3. **Navigation Phase**

   The robot:

   - Turns toward the track
   - Enters line-following mode
   - Continuously adjusts direction based on sensor input

4. **Obstacle Avoidance**

   If an obstacle is detected:

   - The robot executes a predefined avoidance path
   - Then returns to the line

5. **End Detection**

   When all sensors detect black:

   - The robot stops
   - Moves backward
   - Releases the object

Behavior Flow Diagram

```
INPUT (Sensors)
        ↓
Read Distance + Line Sensors
        ↓
PROCESS (State Machine)
        ↓
Decision Making:
  - Follow line?
  - Avoid obstacle?
  - Grab/release?
        ↓
OUTPUT
  - Motors move
  - Servo acts
  - LEDs show status
```

Line-Following Mechanism

The robot uses a weighted sensor system to determine its position:

- Each sensor has a predefined weight
- The system calculates an average position
- The error value represents deviation from the center

Motor speeds are adjusted using:

- Higher error → stronger correction
- Lower error → higher speed

This creates a smooth, proportional control behavior, suitable for curved tracks.

Code Mapping (Design Connection)

The implementation directly reflects the system design:

- Input handling
  - analogRead(SENSOR_PINS[i]) → line detection
  - readDistance() → ultrasonic sensing
- Processing logic
  - switch(state) → state machine
  - lineFollow() → navigation logic
- Output control
  - analogWrite() → motor control
  - servoPulse → gripper control
  - setColor() → LED feedback

Summary

The system is designed to operate on a line-following track, requiring continuous real-time adjustment. By combining multiple sensors, a state machine, and proportional control logic, the robot achieves stable navigation, obstacle handling, and task execution in a dynamic environment.